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Department of Information Technology

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: V Semester B.Tech. IT
Course Code & Title: CCS334 & Big Data Analytics
Name of the Faculty member (s): Mrs.M.Rethina Kumari

Innovative Practice Description

Unit / Topic: Unit 5 / Hive/HBase Command

Course Outcome: CO5

Activity Chosen: Think & response Activity

Justification:

- **Think-&-Response** activity promotes collaborative learning where students think individually, discuss with peers, and share knowledge with the group.
- **Hive and HBase Commands** are crucial in handling data stored in Hadoop. Hive focuses on SQL-like querying, whereas HBase deals with NoSQL storage and retrieval. This activity enables students to learn the commands, their syntax, and practical usage collaboratively, ensuring a deeper understanding of data manipulation and analysis in Hadoop.

Time Allotted for the Activity: 15 minutes

Details of the Implementation:

- The instructor introduced Hive and HBase commands in a classroom session of 30 minutes, covering key commands like creating tables, inserting data, querying in Hive, and performing CRUD operations in HBase. A list of practice commands/questions was distributed to the students.
- **Individual Thinking (1-5 minutes):**

  Students individually think about the given commands, their syntax, and expected outcomes. For instance:

  Hive: CREATE TABLE, INSERT INTO, SELECT

  HBase: CREATE, PUT, SCAN, GET
- **Pair Discussion (6-10 minutes):**

  Students pair up with a peer to discuss their answers. They clarify doubts and share insights on executing the commands effectively. For example:

  - Discuss the differences between Hive and HBase commands.
  - Troubleshoot potential errors in syntax or execution.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO 5	2	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO10**	**PSO1**
	2	1	1	1	1	1
Justification for correlation	It includes logical thinking and make the student able to think about problem solving approach	Able to approach the program by applying what they learnt.	Able to provide solutions for data processing and management using Hive/HBase commands.	Students involved in the activity as team, the team work Skill Improved.	Students' communication skill will be improved as they are facing audience and communicating the activity	Students can use Hive/HBase concepts in software project development for efficient data storage, management, and querying.

- **Images / Screenshot of the practice:**

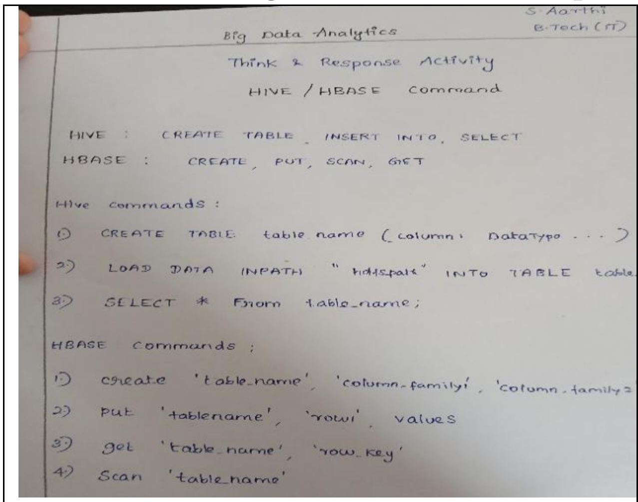
S. Aanthi
B.Tech (IT)
Big Data Analytics
Think & Response Activity
HIVE / HBASE command
HIVE : CREATE TABLE, INSERT INTO, SELECT
HBASE : CREATE, PUT, SCAN, GET
Hive commands :
1) CREATE TABLE table name (column, DataType ...)
2) LOAD DATA INPATH "hdfspath" INTO TABLE table
3) SELECT * From table_name;
HBASE commands :
1) create 'table_name', 'column-family', 'column-family2'
2) put 'tablename', 'row1', values
3) get 'table_name', 'row_key'
4) Scan 'table_name'

Figure. 1a

Figure. 1b

Figure. 1c

Figure. 1d

Glimpse of Think and Response Activity

(Figure.1a shows the list of questions shared to the students, Figure1.b shows team discussion of Devadharshini K, Nandhini S , Kavipraba G and Ananthi P, Figure.1c shows discussion of Sivasekar and Praveen Kumar, Figure.1d shows discussion of Ackshaya M with the class members)

❖ **Feedback of practice from students and other stakeholders:**

- Students found the activity helpful for grasping the practical usage of Hive and HBase commands.
- They reported better retention of command syntax and increased confidence in executing commands.
- Students appreciated the interactive format, which made learning more engaging.

❖ **Benefit of the practice:**

- Improved understanding of Hive and HBase commands, their syntax, and differences.
- Enhanced problem-solving skills through peer discussion.
- Increased confidence in executing and troubleshooting commands in Hadoop.

❖ **Challenges faced in implementation:**

- Some students found it challenging to recall command syntax individually but benefited from peer discussions.
- A few students required additional time to execute and test commands.

References:

1. https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf
2. https://www.readingrockets.org/strategies/think-pair-share
3. https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html
4. https://www.ritrjpm.ac.in/images/B.Tech-IT/20232024/IP/GM_GE3151_U3_Think_pair.pdf

Signature of Faculty Member **HOD**